

AFRITEC LTD

MicroCloud Private Cloud Infrastructure

Troubleshooting Guide / Guide de Dépannage

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Introduction

EN: This document catalogs every technical issue encountered during the AFRITEC MicroCloud deployment, along with the diagnostic steps taken, the root cause identified, and the exact solution applied. These are real-world problems that occurred on production hardware, not theoretical scenarios. If you encounter any of these issues while replicating this deployment, follow the documented resolution.

FR: Ce document catalogue chaque problème technique rencontré lors du déploiement MicroCloud d'AFRITEC, ainsi que les étapes de diagnostic, la cause racine identifiée et la solution exacte appliquée. Ce sont des problèmes réels survenus sur du matériel de production, pas des scénarios théoriques.

Issue 1: SVR02 MRAID Controller Not Visible on PCIe Bus

Symptom / Symptôme

During Ubuntu installation on SVR02, the installer reported 'no disks found.' Despite the CIMC web interface showing all 10 drives as healthy and the RAID virtual drive as Optimal, the Ubuntu live session could not see any storage devices. Running `lspci` showed the MRAID controller was completely absent from the PCIe bus.

Diagnostic Steps / Étapes de Diagnostic

1. Verified all 10 drives show healthy status in CIMC → All Good.
2. Verified Virtual Drive SVR02_OS shows Status: Optimal, Boot Drive: True.
3. Checked CIMC BIOS token: MRAID OptionROM → Enabled.
4. Checked CIMC BIOS token: MRAID PCIe Link Speed → Auto.
5. Performed F9 Optimized Defaults reset in BIOS → No change.
6. Performed Clear BIOS CMOS from CIMC → No change.
7. Ran full PCIe bus scan from live Ubuntu: `lspci -nn | grep '1000:'` → No output. Card completely absent.

Root Cause / Cause Racine

EN: The MRAID mezzanine card was communicating with CIMC over the dedicated management bus (I2C/SPI), which is how CIMC could read drive status and configure RAID. However, the PCIe link from the mezzanine connector to the host CPU was not initializing, making the card invisible to any operating system. This is a physical seating issue — the card was not fully engaged with the mezzanine PCIe connector on the motherboard.

FR: La carte mezzanine MRAID communiquait avec le CIMC via le bus de gestion dédié (I2C/SPI), ce qui permettait au CIMC de lire l'état des disques et de configurer le RAID. Cependant, le lien PCIe du connecteur mezzanine vers le CPU hôte ne s'initialisait pas, rendant la carte invisible pour tout système d'exploitation. C'est un problème d'insertion physique — la carte n'était pas entièrement engagée dans le connecteur PCIe mezzanine sur la carte mère.

Solution

1. Power down SVR02 via CIMC: Host Power > Power Off (graceful).
2. Open the chassis (remove top cover of C220 M5).
3. Locate the MRAID mezzanine card (small card near rear-center of chassis, labeled MRAID or with LSI/Broadcom chip).
4. Press retention clips and lift the card straight out.
5. Reseat firmly — press straight down until it clicks. No rocking, straight down.
6. Close chassis, power on, boot USB, verify with: `lspci -nn | grep '1000:'`

7. Card appeared after reseal. Ubuntu installation proceeded immediately. All RAID/JBOD config survived the reseal.

Key Lesson / Leçon Clé

EN: On Cisco UCS servers, CIMC can communicate with the MRAID card over a separate management bus even when the PCIe link is down. The fact that CIMC shows drives and RAID config does NOT mean the OS can see them. Always verify with `lspci` from a live OS session before assuming storage is available.

Issue 2: Ubuntu Disk Ordering — JBOD Disks Enumerate Before RAID VD

Symptom / Symptôme

On SVR02, after the MRAID reseal, the Ubuntu installer showed the OS virtual drive as sdj (the last device) instead of sda. The JBOD disks occupied sda through sdi.

Root Cause / Cause Racine

Linux enumerates storage devices in the order the controller reports them. When both a RAID virtual drive and JBOD passthrough disks coexist on the same MRAID controller, the JBOD disks are enumerated first because they have lower physical disk IDs. The RAID VD, being a logical construct on top of a physical disk, is presented after all individual disks.

Solution

In the Ubuntu installer, carefully identify the correct disk by its size (approximately 1.14 TB for the RAID VD) rather than its device letter. Use the 'Select drive' dropdown and choose only the RAID virtual drive. Do not use 'Erase all disks' as this would format the JBOD disks intended for Ceph.

Key Lesson / Leçon Clé

Never assume sda is your OS disk on servers with mixed RAID/JBOD configurations. Always verify by size, model name, or serial number. The device letter assignment is non-deterministic and depends on enumeration order.

Issue 3: SVR01 GRUB Bootloader Failure

Symptom / Symptôme

After initial Ubuntu installation on SVR01, the server failed to boot from the OS drive. The GRUB bootloader was broken or not properly installed on the EFI System Partition.

Solution

8. Reflash the USB drive using Rufus with GPT partition scheme and UEFI target system.
9. Boot SVR01 from USB in UEFI mode (F6 > select UEFI: SanDisk).
10. Select 'Try Ubuntu' to enter a live desktop session (not the installer).
11. Install and run boot-repair:

```
sudo add-apt-repository ppa:yannubuntu/boot-repair
sudo apt update && sudo apt install boot-repair -y
boot-repair
```
12. Click 'Recommended Repair' in the boot-repair GUI.
13. After repair completes, remove USB and reboot. GRUB should now load correctly.

Fallback if boot-repair fails:

Manual chroot + grub-install approach from the live session.

Issue 4: Servers Have No Network Connectivity After Ubuntu Install

Symptom / Symptôme

After Ubuntu installation, none of the three servers could reach the network or each other. SVR02 showed 'Cable unplugged' in the network settings. The .52/.53/.54 IPs we were using turned out to be CIMC management IPs, not Ubuntu IPs.

Root Cause / Cause Racine

Cisco UCS C220 M5 servers have a dedicated management port for CIMC that is completely separate from the data plane network interfaces. The CIMC IPs (10.0.1.11/.53/.54) are only accessible through this management port. The Ubuntu operating system uses different physical NIC ports (Port 1 = eno1 on SVR02/SVR03, enp1s0f0 on SVR01) which had no cables connected.

Solution

14. Run physical Ethernet cables from the network switch to Port 1 (eno1) on SVR02 and SVR03. SVR01 uses enp1s0f0.
15. Verify link with `ip a show eno1` (or `enp1s0f0` for SVR01).
16. Assign static IPs via Netplan (see Phase 3 of Installation Guide).
17. Apply with `sudo netplan apply`.
18. Verify all nodes can ping each other.

Issue 5: Snap Install Fails — 'Network is Unreachable'

Symptom / Symptôme

`sudo snap install microcloud --channel=candidate` failed with 'network is unreachable' even though `apt update/upgrade` had worked earlier.

Root Cause / Cause Racine

`apt update` used a local/cached mirror that was accessible on the LAN. Snap's `snapcraft.io` API requires direct internet access. The server's gateway or DNS was not properly configured, or the upstream internet link was temporarily unavailable.

Solution

Verify internet connectivity with `ping -c 3 8.8.8.8`. If it fails, check and fix the Netplan configuration (ensure gateway and nameserver entries are correct). If `snap install` times out but internet is available, simply re-run the command — it resumes the download.

Issue 6: MicroCloud Snap Not Available on 'stable' Channel

Symptom / Symptôme

sudo snap install microcloud failed because MicroCloud was not published to the stable channel.

Solution

Use the candidate channel instead:

```
sudo snap install microcloud --channel=candidate  
sudo snap install microceph --channel=candidate  
sudo snap install microovn --channel=candidate
```

LXD uses latest/stable as normal. Only microcloud, microceph, and microovn require the candidate channel.

Issue 7: SVR01 JBOD Disks Not Visible to MicroCloud/Ceph

Symptom / Symptôme

During microcloud init, SVR01's JBOD disks were not listed as available for Ceph storage. Running lsblk on SVR01 showed combined UCSC-MRAID12G volumes instead of individual disk devices.

Root Cause / Cause Racine

The MRAID controller on SVR01 was not properly exposing the JBOD disks as individual passthrough devices. Instead, it was presenting them as combined controller volumes. This is a CIMC storage policy or firmware-level configuration issue specific to SVR01.

Status / Statut

PENDING: This issue has not yet been resolved. The cluster operates fully with 17 OSDs on SVR02 and SVR03. To fix, access SVR01's CIMC, delete any existing virtual drives on the JBOD disks (if present), and reconfigure them as individual JBOD passthrough. Then add them to the Ceph pool with: `microceph disk add /dev/sdX` (for each disk).

Issue 8: LXD Container Bridge Network Fails in Clustered Mode

Symptom / Symptôme

Running `lxc network create lxdbr0 ipv4.address=10.10.10.1/24 ipv4.nat=true ipv6.address=None` failed because the network had not been declared on each cluster member first.

Root Cause / Cause Racine

In a clustered LXD environment, networks must be created in a two-step process: first declared (with `--target`) on each individual member, then finalized with properties at the cluster level. Additionally, chaining these commands with `&&` failed because the pending state from one target was consumed before the next command could read it.

Solution

Run the target declarations sequentially (one command at a time, wait for each to complete):

```
lxc network create lxdbr0 --target node-01
lxc network create lxdbr0 --target node-02
lxc network create lxdbr0 --target node-03
lxc network create lxdbr0 ipv4.address=10.10.10.1/24 ipv4.nat=true
ipv6.address=None
```

Issue 9: Container DNS Resets on Restart

Symptom / Symptôme

Every time the prometheus or grafana containers were restarted, DNS resolution stopped working. /etc/resolv.conf was reset to empty or an invalid state.

Root Cause / Cause Racine

LXD containers regenerate /etc/resolv.conf on startup based on the network configuration. Since the lxdbr0 bridge was set up without an explicit DNS server, the containers received no DNS configuration.

Solution (Workaround)

After each container restart, manually configure networking:

```
echo "nameserver 8.8.8.8" > /etc/resolv.conf
ip addr add 10.10.10.10/24 dev eth1
ip route add default via 10.10.10.1
```

Permanent fix: Configure the lxdbr0 bridge with dns.domain and dns.mode properties, or set static IP configuration inside the container's /etc/network/interfaces or netplan configuration.

Issue 10: Prometheus YAML Configuration Errors

Symptom / Symptôme

Prometheus failed to start after editing prometheus.yml with nano in the KVM console. YAML parsing errors pointed to incorrect indentation.

Root Cause / Cause Racine

When typing YAML configuration files manually into a KVM HTML console where copy-paste is unavailable, indentation errors are extremely common. A single misplaced space in YAML creates a parsing error that prevents the application from starting.

Solution

Instead of editing with nano (error-prone without copy-paste), use the heredoc approach to write the entire configuration file at once:

```
cat > /etc/prometheus/prometheus.yml << 'EOF'
global:
  scrape_interval: 15s
  evaluation_interval: 15s
scrape_configs:
  - job_name: "prometheus"
    static_configs:
      - targets: ["localhost:9090"]
EOF
```

Tip: Use dot-notation (.) as a mental visualization aid for YAML indentation levels when typing manually: ..key means 2-space indent,key means 4-space indent.

Issue 11: Grafana Binary Name Changed in v10.4.2

Symptom / Symptôme

After downloading and extracting Grafana 10.4.2, running grafana-server (the traditional binary name) returned 'command not found.'

Root Cause / Cause Racine

In Grafana v10.4.x, the binary was renamed from grafana-server to simply grafana. Documentation and tutorials referencing older versions still use the old name.

Solution

```
cd grafana-v10.4.2 && ./bin/grafana server --homepath . &
```

Note the binary is 'grafana' not 'grafana-server', and the 'server' is now a subcommand.

Issue 12: LXD Web UI TLS Token Cannot Be Pasted in KVM Console

Symptom / Symptôme

The LXD web UI at <https://10.0.0.11:8443> requires a TLS trust token for browser certificate authentication. The token is a long alphanumeric string generated on the server. However, the Cisco CIMC HTML KVM console does not support copy-paste from the host workstation, making it impossible to transfer the token to the Windows browser directly.

Root Cause / Cause Racine

EN: The CIMC HTML KVM console operates as a remote framebuffer. While you can type characters, clipboard synchronization between the workstation and the KVM session is not supported. The trust token is too long and complex to type manually without errors.

FR: La console KVM HTML du CIMC fonctionne comme un tampon d'écran distant. La synchronisation du presse-papiers entre le poste de travail et la session KVM n'est pas prise en charge. Le jeton est trop long pour être tapé manuellement sans erreurs.

Solution

Write the token to a file and serve it via Python's built-in HTTP server:

19. Generate the token on SVR01:

```
lxc auth identity create tls/lxd-ui --group admins > /tmp/token.txt
```

20. Serve the file via HTTP:

```
cd /tmp && python3 -m http.server 8080
```

21. On the Windows browser, navigate to <http://10.0.0.11:8080/token.txt> and copy the token.

22. Paste it into the LXD UI authentication page at <https://10.0.0.11:8443>.

23. Kill the HTTP server: `pkill python3`.

Key Lesson / Leçon Clé

Python's built-in HTTP server (`python3 -m http.server`) is an invaluable bridge for transferring text data from a server to a workstation when clipboard and SCP are unavailable. This same technique was reused for Grafana dashboard JSON import.

Issue 13: LXD Containers on OVN Network Cannot Reach the Internet

Symptom / Symptôme

After launching the prometheus and grafana containers, running ping 8.8.8.8 inside either container returned 100% packet loss. The containers had OVN IPs (10.206.37.x) but no route to the internet.

Root Cause / Cause Racine

EN: The OVN UPLINK network was configured for instance traffic within the 10.0.0.150-200 range, but it did not provide NAT-based internet access for containers. OVN manages L2/L3 networking for tenant workloads but does not automatically NAT container traffic to the internet through the host's upstream connection.

FR: Le réseau OVN UPLINK était configuré pour le trafic des instances dans la plage 10.0.0.150-200, mais ne fournissait pas d'accès internet NATé pour les conteneurs. OVN gère le réseau L2/L3 mais ne NAT pas automatiquement le trafic vers internet.

Solution

Create a separate LXD bridge network (lxdbr0) with NAT enabled and attach as a second NIC:

```
lxc network create lxdbr0 ipv4.address=10.10.10.1/24 ipv4.nat=true
ipv6.address=none
lxc network attach lxdbr0 prometheus eth1
lxc network attach lxdbr0 grafana eth1
```

Containers then have two interfaces: eth0 (OVN, cluster traffic) and eth1 (lxdbr0, internet via NAT).

Key Lesson / Leçon Clé

OVN provides software-defined networking for tenant workloads but is not a general-purpose internet gateway. For containers needing to download packages, a separate NAT-enabled bridge is the cleanest solution.

Issue 14: Grafana Dashboard Import Fails — Cannot Reach grafana.com

Symptom / Symptôme

In the Grafana web UI, entering dashboard ID 1860 and clicking Load returned an error. Grafana could not reach grafana.com to download the dashboard JSON because the Grafana container had no direct internet access.

Root Cause / Cause Racine

EN: The Grafana dashboard import feature requires the Grafana server process itself (not the user's browser) to make an outbound HTTP request to the Grafana.com API. Even with lxdbr0 configured, Grafana's server-side HTTP client could not resolve grafana.com due to container DNS limitations.

FR: La fonctionnalité d'importation nécessite que le processus serveur Grafana lui-même fasse une requête HTTP sortante vers l'API grafana.com. Même avec lxdbr0, le client HTTP côté serveur de Grafana ne pouvait pas résoudre grafana.com.

Solution

Download the dashboard JSON on a host with internet access and serve it locally:

24. On SVR01 (host):

```
wget https://grafana.com/api/dashboards/1860/revisions/37/download -O /tmp/node-exporter-dashboard.json  
cd /tmp && python3 -m http.server 8080
```

25. On the workstation browser, open <http://10.0.0.11:8080/node-exporter-dashboard.json> and copy the entire JSON (Ctrl+A, Ctrl+C).

26. In Grafana: Dashboards > New > Import > paste the JSON in the JSON model box.

27. Select Prometheus as the data source and click Import.

Key Lesson / Leçon Clé

In internet-restricted environments, pre-download all external dependencies on a host with internet access and transfer locally. The Python HTTP server technique works for any file transfer where SCP or copy-paste is unavailable.

Issue 15: Prometheus Container Cannot Scrape Host Node Exporter

Symptom / Symptôme

After installing Node Exporter on SVR01's host OS (port 9100) and configuring Prometheus to scrape 10.0.0.11:9100, the Prometheus targets page initially showed the endpoint as DOWN.

Root Cause / Cause Racine

EN: The Prometheus container's primary network interface (eth0) was on the OVN internal network (10.206.37.x). This network segment does not have a route to the host's physical network (192.168.100.x). The container needed a second interface on the lxdbr0 bridge that provides NAT connectivity to the physical network.

FR: L'interface réseau principale du conteneur Prometheus (éth0) était sur le réseau interne OVN (10.206.37.x). Ce segment n'a pas de route vers le réseau physique de l'hôte (192.168.100.x).

Solution

After attaching lxdbr0 to the Prometheus container and configuring eth1, verify connectivity first:

```
# Inside prometheus container:
curl http://10.0.0.11:9100/metrics
```

If metrics data is returned, update prometheus.yml to target host IPs:

```
- job_name: 'node_exporter'
  static_configs:
    - targets: ['10.0.0.11:9100', '10.0.0.12:9100', '10.0.0.13:9100']
```

Key Lesson / Leçon Clé

Containers on OVN and hosts on the physical network are on different L3 segments. For monitoring containers to scrape host metrics, they need a network path to the physical network through NAT. Always verify with curl before assuming Prometheus config is correct.